

# Undavalli v surya prakash

Third Year Undergraduate

Department of CSE, SRM UNIVERSITY,kattankulathur

uvsuryaparakash17@gmail.com

+91 9391551617

## Education

| Degree    | Institution              | CPI/% | Year           |
|-----------|--------------------------|-------|----------------|
| B.Tech    | SRM University           | 9.05  | 2023 – Present |
| Class XII | Tirumala Junior Kalasala | 96.6  | 2021 – 2023    |

## Projects

### Calorie Burnt Prediction

- Built and deployed a predictive model using **XGBoost** to accurately forecast calories burnt with **97% prediction accuracy** based on activity level, duration, and individual biometric characteristics.
- Applied advanced machine learning algorithms to analyze **10,000+ real-world health and fitness records**, improving model generalization.
- Utilized statistical methods (**cross-validation, RMSE, MAE**) resulting in a **15% improvement** over baseline regression models.
- Reduced prediction error by **12%** and improved model training efficiency by **20%** through feature engineering and optimization.

### Audio Deep Fake Prediction

- Developed a robust **Convolutional Neural Network (CNN)** to detect synthetic audio attacks, achieving an **AUC of 0.945** and **90% accuracy** on unseen data.
- Engineered a preprocessing pipeline converting raw audio into **Mel Spectrograms** and handled severe **9:1 dataset imbalance** using Random Up-sampling.
- Implemented **Spec Augment** (time and frequency masking) to mitigate overfitting and generalize across **13 unseen spoofing algorithms (A07A19)**.
- Achieved an **Equal Error Rate (EER) of 9.43%**, validating reliable detection of advanced voice clones.

## Skill Summary

- **Languages:** Python, Verilog
- **Tools:** MySQL, MATLAB, Simulink, PSpice
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn
- **Coursework:** Data Structures & Algorithms (DSA), OOPS, SQL
- **Areas of Interest:** Machine Learning, Data Analysis
- **Soft Skills:** Problem Solving, Project Management, Team Collaboration, Self-learning, Presentation

## Extra-Curricular Activities

- Cricket & Badminton